Project and Professionalism (6CS007)

Literature Review

Submitted by: Chirag Lama

Student ID: 2434743

Supervisor: Ramesh Malasi

Reader: Nerisha Shrestha

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1. Initial Research (Literature Review)

1.1. Literature Review #1 - Resume Analysis Using NLP and ATS Algorithms (Saurabh Yadav, 2025)

The examined literature emphasizes notable technological progress in automated resume screening via the incorporation of Natural Language Processing (NLP), Machine Learning (ML), and Applicant Tracking Systems (ATS). Research consistently shows that conventional manual screening is lengthy, variable, and susceptible to human mistakes. Researchers have created advanced computational models to enhance the speed, efficiency, and accuracy of resume assessment.

Initial methods concentrated on fundamental machine learning classifiers like KNN, SVM, and XGBoost, which were employed to forecast job roles and classify resumes. Subsequent studies presented deep learning frameworks, such as CNNs, LSTMs, and multitask models intended for resume segmentation and entity extraction, greatly improving parsing precision. Numerous studies highlight the significance of Named Entity Recognition (NER) in obtaining essential information—like skills, experience, education, and job title from semi-structured or unstructured resume formats.

Recent research indicates a move towards BERT-oriented and transformer-powered NLP frameworks, showcasing enhanced effectiveness in feature extraction and similarity assessment between resumes and job postings. Ensemble models and layout-sensitive systems enhance precision by identifying structural elements of resumes, including headings, sections, and formats (PDF, Word, etc.). Studies also emphasize the importance of recommendation methods and matching algorithms that evaluate candidates according to their fit for the job, frequently utilizing cosine similarity, keyword matching, or decision-tree techniques.

Moreover, the literature highlights increasing worries regarding fairness and the perception of automated systems. Certain research indicates that although ATS tools improve efficiency, both users and applicants occasionally view them as more biased or less transparent compared to human assessors, highlighting the demand for more understandable AI-based solutions.

The body of research collectively shows a distinct path toward increasingly sophisticated, precise, and user-focused resume analysis systems, incorporating AI-based parsing, relevance scoring, job role prediction, and automated job-fit evaluation.

Findings:

- NLP-based resume parsing significantly improves the extraction of key information such as skills, roles, contact data, and education.
- Deep learning and transformer models (e.g., CNN, LSTM, BERT) outperform classical ML models in predicting job roles and matching resumes to job descriptions.
- Layout-aware and multitask resume parsers reduce parsing time and increase accuracy for multi-format resumes.
- Automated ATS scoring enhances objectivity but raises potential fairness concerns in user perception.
- Ensemble and stacked models provide more reliable candidate ranking compared to single-model approaches.
- AI-driven systems offer personalized feedback, resume optimization guidance, and improved candidate-job alignment.

1.2. Literature Review #2 – Resume Screening with Natural Language Processing (Mehtap Saatçı[1], 2024)

The purpose of this research is to examine how organizations conduct manual screenings of thousands of resumes, resulting in inefficiency, bias, and missed chances to accurately select candidates. The authors suggest using an automatic resume screening system built with Natural Language Processing (NLP) methods to find competencies through the extraction process, similar to those predetermined skill sets for specific job positions. When examining various similarity metrics, Jaccard Similarity and Cosine Similarity, the authors found that Jaccard's approach produced better results for competency-based matching. A case study looking at 123 different job positions and reviewing 16 resumes was used to illustrate how an organization can effectively use this type of system to rank candidates' suitability for positions, allowing the organization to spend less time on the screening process, experience reduced levels of bias, and increase accuracy when selecting candidates. The use of LinkedIn had additional benefits because it allowed users to upload their resumes with ease. Furthermore, the authors state that this system has the potential for scalability in many different industries and emphasizes the importance of ethical standards when conducting automatic resume screenings; these include

upholding principles of transparency and fairness. Importance of ethical standards when conducting automatic resume screenings; these include upholding principles of transparency and fairness.

Findings:

- Manual resume screening is inefficient, error-prone, and susceptible to bias.
- NLP-based systems can automate competency extraction and candidate ranking, improving fairness and efficiency.
- Jaccard Similarity provides more accurate results than Cosine Similarity for skill-based matching.
- The system demonstrates scalability across technical and managerial roles, with potential for broader sectoral application.
- Ethical concerns (bias mitigation, transparency, data privacy) are critical for adoption of AI-driven recruitment tools.

1.3. Literature Review #3 – Resume Optimisation and Suggestions using Large Language Models (Sai Ganesh Mandala1, 2024)

This article will evaluate the ways in which large language models (LLMs) can improve the format of a CV (Curriculum Vitae) so that it meets the needs of an employer. The authors review previously published research on the use of artificial intelligence (AI) applications to optimize CVs, and discuss how LLMs create suggested content that is geographically relevant, enhances language, and emphasises pertaining job role skills that can ultimately maximize compatibility with applicant tracking systems (ATS).

The authors discuss LLM use as both: 1) an LLM tool for a CV's development and enhancement, and 2) an LLM evaluator that mimics the perspective of an employer or recruiter.

The authors additionally identify several of the more difficult challenges that LLMs potentially face, including issues related to bias and ethics, and their potential risk of creating over-standardisation - whereby the voice of the individual is removed from the CV.

In conclusion, the authors frame LLM tools as intermediaries between companies (recruiters) and job seekers, and provide scalable but highly-customized support, yet at the same time raises questions regarding both the fairness of, and level of transparency associated with, automated career tools.

Findings:

- LLMs can significantly improve resume structure, clarity, and ATS compatibility.
- AI-driven suggestions help job seekers highlight relevant competencies and tailor resumes to specific roles.
- Ethical concerns include bias, fairness, and the potential homogenization of resumes.
- LLMs act both as **assistive generators** and **evaluators**, simulating recruiter expectations.
- The technology demonstrates promise for scalable, personalized career support but requires careful governance.

1.4. Literature Revie #4 – AI-Powered Resume Generator: Enhancing Employability through Intelligent Design (Dr. Y Sagar1, 2024)

This academic paper presents an evaluation of the impact that AI (artificial intelligence) has had on how resumes are created and optimized for sourceable content, and what that will look like for job seekers and those who evaluate resumes. Using Natural Language Processing (NLP) technology along with Machine Learning (ML), AI has enabled the development of Resume Generators, which use NLP/ML capabilities to create optimized resumes based on the specific job description and ATS.

The study finds that AI Resume Generators offer the following benefits to job seekers as well as recruiters: Individualized Resume format which gives them unique profiles, data-driven recommendations to help them tailor their resumes for a particular position, and standardization of the resume format to assist Recruiters to evaluate candidate's qualifications more easily.

Also, the paper highlights several challenges associated with using AI Resume Generators, such as the ethical concerns about using algorithms that incorporate human thought processes, potential bias in the algorithmic design, and potential loss of individuality due to the standardization of resumes.

As a whole, this research shows that AI Resume Generators represent a critical new 'bridge' between candidates and employers, and act to promote increased employability within increasingly competitive labor markets.

Findings:

- AI resume generators improve employability by tailoring resumes to job descriptions and ATS requirements.
- NLP and machine learning enable personalized, competency-focused resume suggestions.
- These tools reduce recruiter workload by standardizing resume formats and highlighting relevant skills.
- Ethical challenges include bias, fairness, and the potential loss of individuality in resumes.
- AI resume generators act as a bridge between job seekers and recruiters, fostering efficiency and accessibility.

1.5. Literature Review #5 - AI-Powered Resume Screening and Ranking System (AlbassamA, 2023)

The aim of this research is to develop a computer-based resume screening and ranking tool that provides a solution for the inefficiencies and discrimination that characterise traditional recruitment methods. The authors put forward a framework that combines Natural Language Processing (NLP), machine learning and Applicant Tracking System (ATS) principles to automate the assessment of candidates. The tool extracts competencies from resumes, assesses them in relation to job requirements and ranks them according to their suitability score. A case study is presented to demonstrate how using the tool reduces the workload of recruiters while improving fairness and accuracy in selecting candidates. The paper also discusses the potential for scalability to other industries and addresses the ethical considerations that arise, such as transparency, bias mitigation and data privacy. Through combining algorithmic knowledge and the practical recruitment requirements, the authors propose that AI-based resume screening is a disruptive phenomenon in modern recruitment.

Findings:

- Manual resume screening is inefficient and prone to bias, necessitating AI-driven solutions.
- NLP and machine learning enable competency extraction and candidate ranking aligned with job requirements.
- The system improves recruiter efficiency by automating evaluation and reducing human error.
- Ethical challenges include fairness, transparency, and privacy concerns in AI-based recruitment.
- Demonstrates scalability across diverse industries, making AI-powered screening broadly applicable.

1.6. Literature Review #6 - AI-Powered Resume Screening: A Comparative Study of Algorithms and Approaches (Kriebitz1, 2023)

This paper examines how AI can improve recruitment by comparing several AI algorithms' abilities to automate the screening process and evaluate candidates through the lens of resume screening. The authors focus on three main types of AI techniques- natural language processing (NLP), machine-learning-based classifiers, and semantic similarity measures-and compare each one's effectiveness in accurately and fairly matching candidates with job postings. While traditional resume screening using keywords often overlook more nuanced qualifications, more complex AI algorithms enable users to consider more than just the keywords on a resume; rather, they can relationally match the candidate's skills with the job needs, as well as utilize the candidate's overall competencies. In evaluating different techniques for automating candidate screening and evaluation, the authors also discuss the need for organizations to find equilibrium between speed, accuracy, and fairness when selecting which AI tools to use during the recruitment process.

Findings:

- Traditional keyword-based screening is limited and prone to overlooking contextual skills.

- Advanced AI models (NLP + semantic similarity) improve accuracy in candidate-job matching.
- Comparative analysis shows trade-offs between efficiency, fairness, and interpretability across algorithms.
- Algorithmic bias remains a significant concern, requiring transparent design and ethical safeguards.
- Organizations must align algorithm choice with recruitment priorities (speed vs. fairness vs. accuracy).

1.7. Literature Review #7 – AI-Powered Resume Screening and Candidate Matching Framework (Luetge, 2022)

The authors of this paper provide a detailed description of a holistic approach towards AI-powered resume screening and candidate-job matching. Their use of Natural Language Processing (NLP), semantic similarity methods and machine learning systems promotes an automated evaluation process for resumes matched to job descriptions, as compared with traditionally keyword-driven methodologies. The model focuses primarily on the understanding of the context of skills, experience and competencies to allow for a greater degree of accuracy when selecting candidates. The results of the case study indicate that the model can effectively rank applicants, therefore lessening the burden on recruiters while at the same time greatly reducing human bias. This paper also articulates the benefits of an AI recruitment tool across a wide array of sectors, and discusses the ethical implications associated with fairness, transparency and privacy for individuals using AI recruitment tools. This research paper illustrates how AI can fundamentally change the manner in which recruitment is conducted through a combination of research and practical examples.

Findings:

- Traditional keyword-based screening is limited; contextual AI models improve candidate-job matching.
- NLP and semantic similarity algorithms enhance accuracy in competency evaluation.
- Automated ranking reduces recruiter workload and mitigates bias in selection.

- Ethical challenges include fairness, transparency, and privacy concerns.
- The framework demonstrates scalability across diverse industries and job roles.

1.8. Literature Review #8 – conSultantBERT: Fine-tuned Siamese Sentence-BERT for Matching Jobs and Job Seekers (Dor Lavi, 2021)

The present study demonstrates a new tool called conSultantBERT which is based on a state-of-the-art model for similarity matching utilised in hiring people called Siamese Sentence-BERT (SBERT). This tool was created to assist employers by more accurately finding candidates who meet company-specific hiring criteria via creating embeddings of resume text and job posting text. The authors point out various challenges associated with this area of research: noisy resume data; different formats for various documents; and the requirement of creating an accurate training dataset that contains cross-linguistic and multilingual content in a global job market. To accomplish these objectives, the researchers developed their own large training dataset of over 270,000 resume-job opportunity pairs with the assistance of staffing consultants. The end result indicates that the conSultantBERT model outperforms any unsupervised baseline method (TF-IDF, pre-trained BERT) as well as supervised baseline methods (Random Forest Classifiers). The experimental results show that the conSultantBERT generated superior ROC-AUC, Precision, Recall and F1 Scores. Additionally, the researchers were able to generate high-quality embeddings using regression-based Cosine Similarity which had the highest performance within the conSultantBERT framework. Another important attribute of the conSultantBERT framework is its ability to correctly match an English resume to a Dutch posting, or vice versa, which has been a major barrier for businesses wishing to successfully operate in the multinational environment.

Findings:

- Traditional unsupervised methods (TF-IDF, pre-trained BERT) perform poorly in resume–vacancy matching.
- Supervised baselines improve performance, but fine-tuned conSultantBERT embeddings outperform both.

- Regression-based cosine similarity optimization yields more robust embeddings than classification objectives.
- The model effectively handles noisy, heterogeneous resume data and structured vacancy texts.
- Multilingual and cross-lingual capabilities make conSultantBERT scalable for global recruitment systems.

1.9. Literature Review #9 – National Origin Discrimination in Deep-Learning-Powered Automated Resume Screening (Sihang Li, 2023)

The purpose of this academic research paper is to evaluate the ethical and legal implications of using automated resume screening systems powered by deep learning in relation to discrimination against individuals based on their nationality. The authors demonstrate that the use of word embedding algorithms in NLP for screening resumes can reflect biases present in the corpora used to train the algorithms (e.g. Google News, Wikipedia) from which the algorithms learned. These word embeddings reflect socio-linguistic behaviour of the authors, which may consequently accidentally encode demographic information regarding nationality, ethnic group and culture. Thus, automated resume screening systems can favour or disfavour candidates based on their nationality, which could be a violation of equal opportunity laws (i.e. GDPR, CCPA, EEOC). The article provides several potential solutions for bias mitigation, including the use of fairness metrics and creating adjusted word embeddings that give lower weight to bias-prone terms. The experimental results provide support for the assertion that national origin bias exists and that bias mitigation strategies are useful in overcoming this issue.

Findings:

2. Deep learning-based resume screening improves efficiency but inherits bias from training corpora.
3. Word embeddings encode demographic signals (e.g., nationality, ethnicity), leading to disparate impacts.
4. National origin discrimination occurs when resumes from certain groups are systematically favored/disfavored.

5. Fairness measures and bias mitigation techniques (e.g., adjusted embeddings, sigmoid weighting) can reduce discrimination.
6. Ethical and legal implications are significant, requiring transparency, fairness audits, and regulatory compliance.

1.10. Literature Review #11 – The Role of Artificial Intelligence in Recruitment Process Decision-Making (Adel Ismail Al-Alawi, 2021)

Through scholarly research, this article highlights how Artificial Intelligence (AI) is changing the recruitment and selection process by being able to assist and enhance recruitment through their application of AI tools, addressing barriers in recruiting through efficiencies and reducing discrimination in recruitment. The research looks at multiple years of previous research from 1988-2020 and highlights areas where AI has benefited recruiting in resume screening, candidate sourcing, psychometric testing, video interviewing, monitoring employer branding and background checks. Although the authors state there are increasing numbers of companies, especially larger high-tech firms, that are incorporating AI into their recruiting process, interviewing candidates will continue to be a key component of the recruiting process and thus, relative to continuing issues of human bias in recruitment, AI cannot provide a complete solution to eliminate biases. The authors also highlight that when adopting AI, these new technologies create ethical challenges regarding fairness, diversity, and automated bias; therefore, while automating recruiting through AI will reduce the level of discrimination in recruiting, AI cannot completely eliminate the biases caused by cultures and/or motivations. The case studies included in this paper, particularly IBM's Watson Candidate Assistant, highlight benefits of improved candidate-to-job fits, as well as decreased time-to-hire, of using AI tools. Other AI vendors mentioned as significant participants in AI recruitment include Workday, BambooHR, Monster Talent Management, and "Google for Jobs".

Findings:

- AI addresses inefficiencies in traditional recruitment, including costly resume reviews and candidate sourcing.

- Tools such as vacancy prediction, job description optimization, and CV screening enhance fairness and efficiency.
- AI adoption is concentrated in large, high-tech firms, with slower uptake in smaller organizations and public sectors.
- Ethical challenges include transparency, cybersecurity risks, and cultural barriers in AI-driven recruitment.
- Benefits include reduced administrative workload, faster recruitment, improved candidate-job matching, and enhanced competitiveness.
- Human bias persists, as interviews and motivational assessments remain outside AI's scope.

2. Summary

This report highlights the increasing impact AI is having on the recruitment process and the screening of resumes as it relates to technological advancements and ethical implications. In many studies, AI technologies including NLP, LLMs and deep learning embeddings have been used to increase efficiency, accuracy and scalability in matching candidates with jobs. For example, conSultantBERT has demonstrated improved performance in both multilingual and cross-lingual environments. In addition, AI powered resume generators and optimizers have helped job seekers create more employable resumes through ATS.

However, the ethical issues surrounding the potential for bias, fairness, transparency, and diversity are significant. Multiple studies indicate that there is potential for national origin discrimination, gender bias, and ethical dilemmas when algorithms learn and build from the stereotypes embedded in training data. Vendors and companies using AI tools (for example, IBM, Google, LinkedIn) highlight the practical advantages of employing AI, which include reduced time to hire, improved candidate engagement, but they still acknowledge that companies need to have human oversight over the recruitment process to avoid bias and assist in finding candidates who will be successful within the company's culture.

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